Mask2Map: Vectorized HD Map Construction from BEV Segmentation Masks

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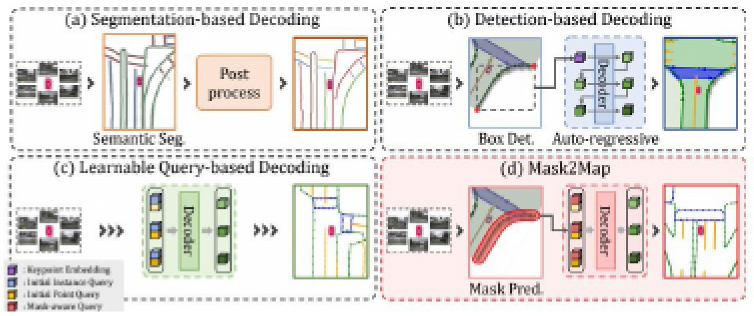
02

Context & Motivation for HD Maps

3 Why HD Maps Matter for Online Autonomy

- HD maps enable precise localization, planning, and safety.
- Offline construction is complex, costly, and slow to update.
- Onboard mapping faces latency, memory, robustness limits.
  - Online methods use onboard sensors to reflect conditions.
  - Segmentation lacks precision at key sites, hurting forecasting.
- Vectorized maps via segmentation, detection, query decoding.

4 Mask2Map: Decoding Paradigms Compared



- Segmentation: dense masks, weak instance granularity
  - Detection: discrete boxes miss fine topology cues
  - Learnable queries: flexible but query design heavy
- Mask2Map: BEV masks steer instance-level decoding
  - IMPNet captures global semantics from BEV masks
  - MMPNet refines local geometry into vector maps

Sec 5

Related Works Landscape

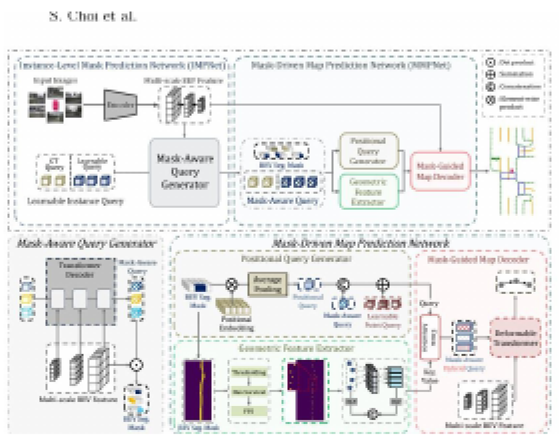
Positioning Mask2Map Among Prior Methods

- LSS and BEVFormer uplift 3D BEV features and segmentation.
- HDMapNet and VectorMapNet vectorize maps from BEV features.
- Two decoding lines: polyline extraction vs query-based.
  - Real-time sensor cues enable detailed, instance-level maps.
  - DN-DETR injects noisy GT to stabilize query-GT matching.
- These ideas inspire Mask2Map's inter-network denoising.

Sec 3

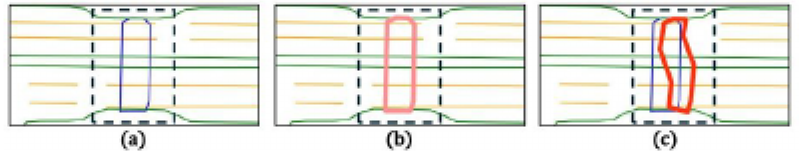
Mask2Map Method Overview

IMPNet and MMPNet Architecture Details



- BEV Encoder fuses camera–LiDAR into multi-scale features
  - Mask-Aware Query Generator yields BEV mask queries
  - Positional Query Generator supplies spatial priors
- Geometric Feature Extractor refines map geometry
  - Query–mask interactions via dot-product and concatenation
  - Mask-Guided Map Decoder outputs vectorized elements

9 Inter-network Denoising Training

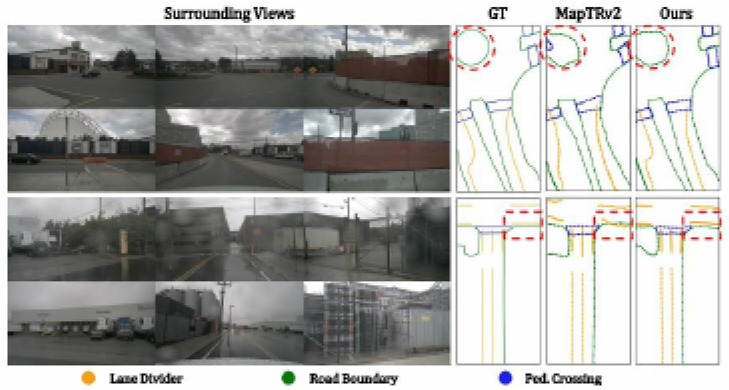


- Injects controlled noise between IMPNet–MMPNet
  - Uses noisy GT masks to stabilize query matching
  - Variants (a–c) regularize and diversify training
- Improves robustness to imperfect segmentation cues
  - Enhances topology consistency across map elements
  - Yields higher accuracy with smoother convergence

Sec 4

Experiments & Results

11 Quantitative & Qualitative Results



- Datasets: nuScenes and Argoverse2 benchmarks.
  - Metrics: AP via Chamfer and rasterization.
  - Strong mAP gains vs MapTRv2 and others.
- Works with camera-only and camera-LiDAR fusion.
  - Ablation: IMPNet, MMPNet boost map accuracy.
  - Inter-network denoising further improves mAP.

12 Discussion, Limitations, and Conclusion

- Limitations: edge cases and domain shifts remain challenging.
- Temporal cues planned to resolve occlusions across frames.
- Model optimization to boost FPS without accuracy loss.
  - Broaden semantic categories to enrich map expressiveness.
  - Leverage self-supervision to reduce label reliance.
- Conclusion: Global+local semantics + IDT deliver SOTA results.